2026 MIDDLE SCHOOL STEM CAMPS

From Trees to Tech

INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

IN THIS PACKET

MudWatt Bioelectric League

Turn a cup of mud into a living battery

At a glance: Can your team make mud generate electricity and defend your design with data? About 80 min; 4 teams.

MATERIALS

- 4 DIY soil microbial fuel cells: clear deli container, pond mud, two 4×4 in carbon-felt electrodes each (plus 1 spare set)
- measured sugar cubes as fuel (one per cell per run, plus spares)
- 4 multimeters, one per team, plus 2 spare (confirm battery type and that a battery is included)
- 100 kΩ resistors and alligator-clip lead wires, one set per cell
- nitrile gloves and a trash-bag drip liner per station
- labels and masking tape
- handwashing supplies

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage reading	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**

SOURCES soil microbial fuel cell electrode and voltage references (carbon-felt anodes; terrestrial MFC literature on 300 to 600 mV output as the biofilm matures).

Daily voltage log

Each day, set the multimeter to DC millivolts and read the voltage across the 100 kΩ resistor at the same time; the cell is weak on day 1 and climbs as the biofilm grows. Defend the design with the trend, not one number.

Team ____________________ The ONE variable we are testing ____________________

DAY	TIME	VOLTAGE (mV)	NOTES
Mon Jun 22			
Tue Jun 23			
Wed Jun 24			
Thu Jun 25			
Fri Jun 26			
Peak reading			**Day of peak**

Forest Sensor Sprint

Site the smartest place for a campus tree sensor

At a glance: Find the smartest place to install a future campus tree sensor. About 80 min; 4 teams.

MATERIALS

- 4 clipboards plus spares
- 4 paper clinometers plus spares
- 4 combined temperature + humidity meters (one device, plus spares)
- 4 three-in-one soil meters (moisture, light, pH), one per team, plus spares
- 1 to 2 long tape measures (pre-mark a fixed standoff distance in prep)
- 4 route cards (one per team) plus 2 spares
- pencils

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: calibrate against the reference first, record every reading with its units, and plan a route with no backtracking.
- Calibrate together, then measure temperature, humidity, light, and soil moisture with units at each checkpoint and estimate the marked tree's height.
- Order the checkpoints to minimize backtracking, then recommend one sensor site and defend it with your own numbers.
- Score, debrief with evidence, reset the station.

DEBRIEF

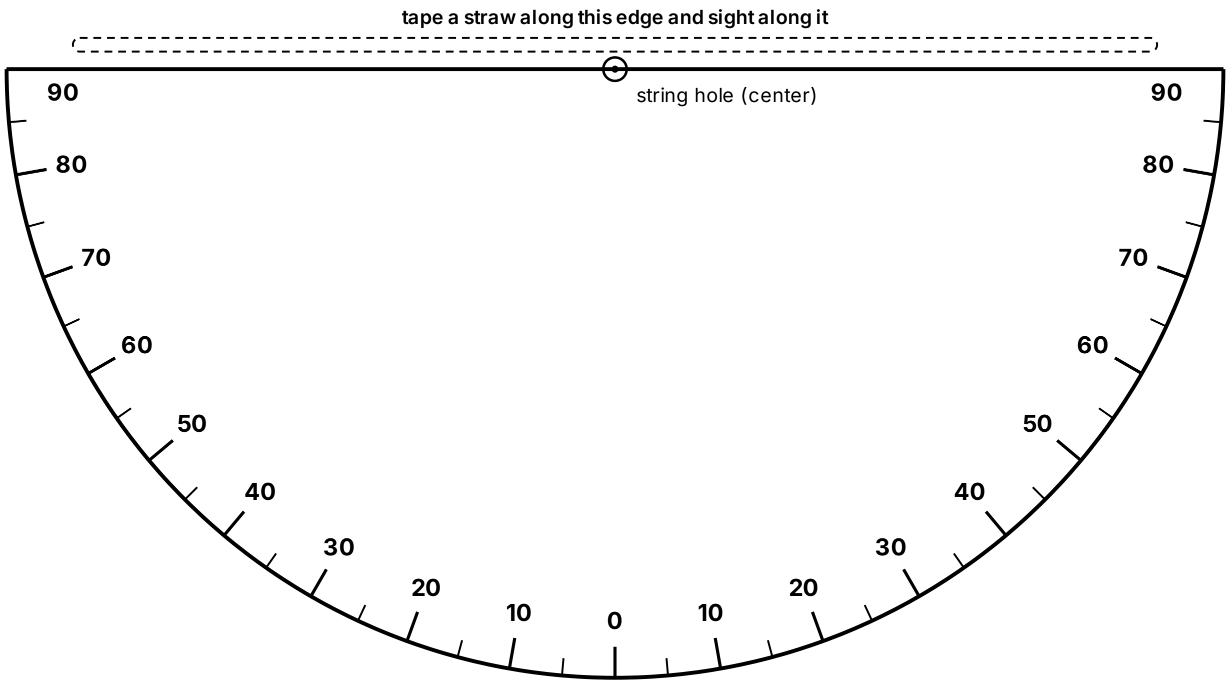
- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	**100**

SOURCES GLOBE Observer Trees; NASA MyNASAData Urban Heat Island

Paper clinometer



Cut along the flat top edge and the curved outline.

ASSEMBLE

01 Cut out the half circle along the curved outline.

02 Tape a straight straw along the top straight edge.

03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.

04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.

05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor marked.

ANGLE	TAN
10	0.18
20	0.36
30	0.58
40	0.84
45	1.00
50	1.19
60	1.73
70	2.75

Example: eye height 1.5 m, distance 10 m, angle 40° gives 1.5 + 10 × 0.84 = about 9.9 m.

Team route card

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ________________________ Date ______________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: ______________________________________

Why (use your data): ______________________________________

The checkpoints are the spots the instructor marked outside.

cut here

From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team ________________________ Date ______________

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

CHECKPOINT	VISIT ORDER	TEMP (°F)	HUMIDITY (%)	LIGHT (RELATIVE)	SOIL MOISTURE (1 TO 10)	NOTES
Reference (calibrate)						
A						
B						
C						
D						
E						

Our recommended sensor site: ______________________________________

Why (use your data): ______________________________________

The checkpoints are the spots the instructor marked outside.

Standoff floor marker

Place it at the distance you pre-mark from the tree base with the long tape.

FROM TREES TO TECH 2026 · TTT-02

Tree-height station

Stand here

Sight the treetop through your clinometer straw from this mark.

This mark is a measured distance from the tree base. Write that distance here: __________ m, then use it in your tree-height formula.

Seed Dispersal Derby

Engineer a seed that escapes its parent tree

At a glance: Build a seed that escapes the parent tree without engines or throwing. About 80 min; 4 teams.

MATERIALS

- paper
- cardstock
- paper clips
- masking tape
- string
- fan or marked test lane
- a shared tape measure for distance
- timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
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- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which change improved hang time without losing distance?
- Why does spinning slow the fall?
- Which tree's strategy did your final seed copy?

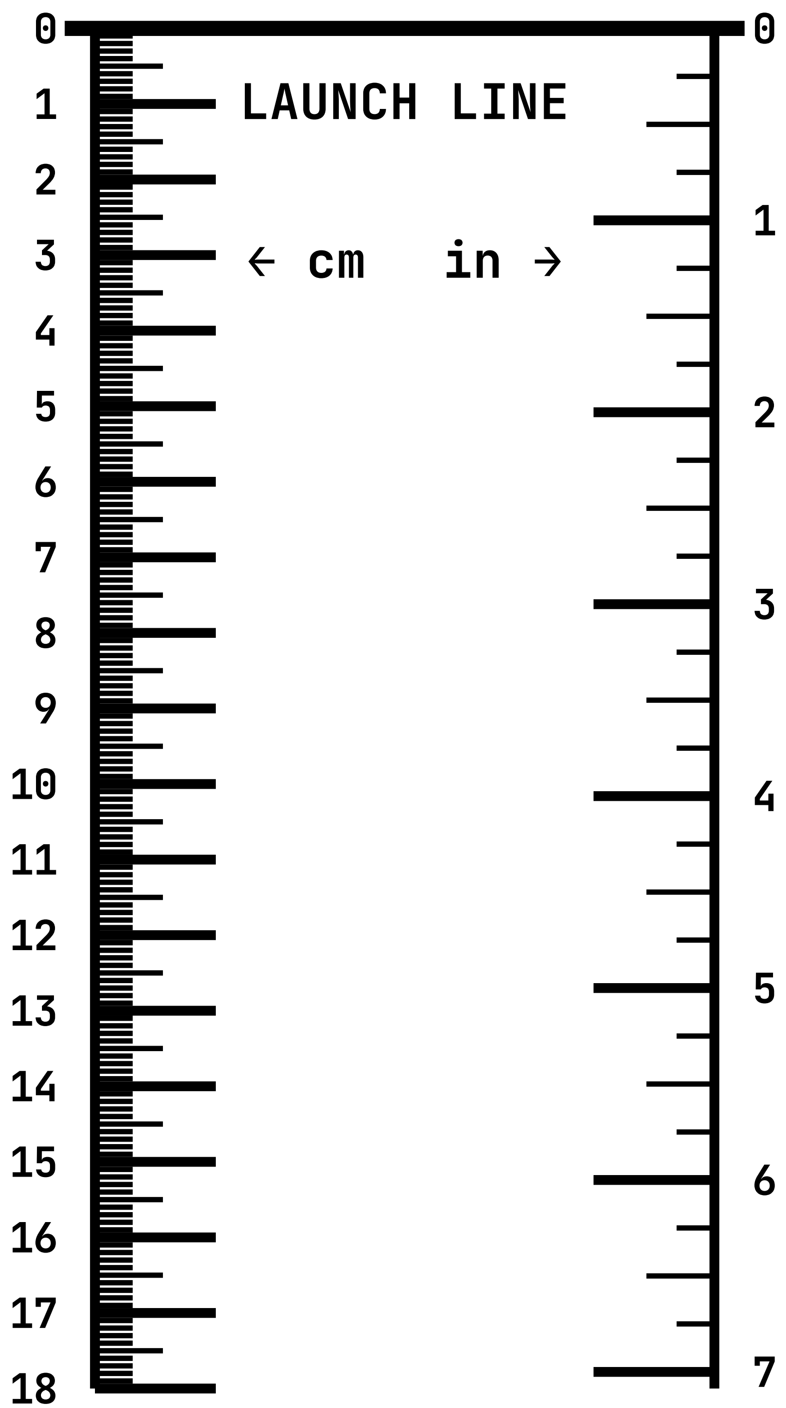
SCORING (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	**100**

SOURCES Science Buddies Seed Dispersal; Project Learning Tree

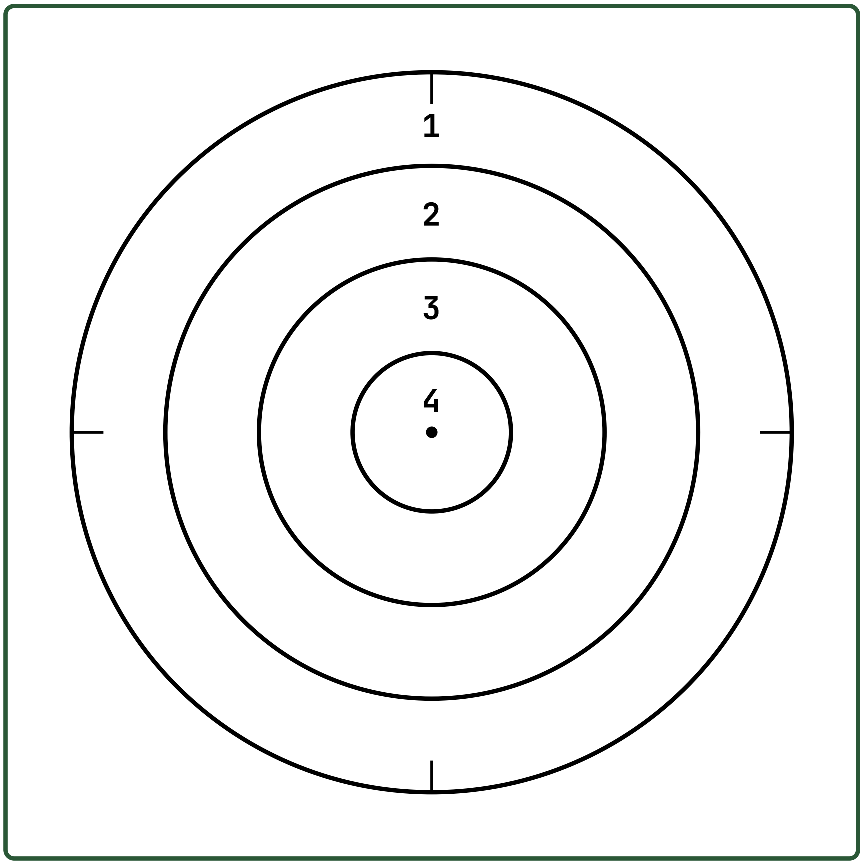
Drop-lane distance strip

Tape the strip flat down the lane with the **0** end at the launch line.



Landing-zone target

Lay the target flat where staff pick down the lane and line the crosshairs up with the lane center. A higher ring number is a better landing.



Ring 1 Outer ring. Your seed touched the target.	**Ring 2** Good aim. Getting closer.	**Ring 3** Close to the center.	**Ring 4** Bullseye, right on the dot.

Xylem Pipeline Relay

Move water uphill the way a tree does

At a glance: Move water uphill like a tree, faster and cleaner than rival teams. About 80 min; 4 teams.

MATERIALS

- paper towels
- cotton balls
- cotton cloth strips
- cups
- food coloring
- trash bag table cover
- rulers
- elevation blocks

FACILITATION ARC

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- Score, debrief with evidence, reset the station.

DEBRIEF

- Which material moved water best, and why?
- How did slope change your delivery?
- Where does a real tree use this same effect?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	**100**

SOURCES Science Buddies; plant water transport references

Greenhouse Climate Controller

Match plants to the right climate zone

At a glance: Run the greenhouse controls for a plant that cannot complain, only wilt. About 80 min; 4 teams.

MATERIALS

- plant profile cards
- control dial boards
- clipboards
- dry erase markers
- tour clue sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: match on what each plant needs, use the tour evidence, and change one placement at a time.
- Sort the plant cards by need, match each plant to a zone, and set temperature, humidity, and light on the dial board using the tour evidence.
- Where plants conflict in one zone, pick the setting your evidence best supports and note the tradeoff.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	**100**

SOURCES Temple Ambler Greenhouse

Plant profile cards

Each card says what one plant needs. Match it to the greenhouse zone whose tour readings fit, then defend the placement with a clue from the tour.

Fern

TEMP	cool to warm, 16 to 24 °C (60 to 75 °F)
HUMIDITY	high, 50% or more; likes damp air
LIGHT	shade or bright indirect; never direct sun

WANTS Damp air and shade, like a forest floor.
AVOID Direct sun and dry air crisp the fronds.

Moth orchid

TEMP	warm, 24 to 29 °C (75 to 85 °F) by day, above 16 °C (60 °F) at night
HUMIDITY	humid, 50 to 80%, with some air movement
LIGHT	medium; bright but filtered, no direct beam

WANTS Warm, humid air and bright, filtered light.
AVOID Full direct sun scorches the leaves (the orchid-loves-sun myth).

Cactus

TEMP	warm days, big day-to-night swing, 24 to 34 °C (75 to 90 °F)
HUMIDITY	low; dry air and fast-draining soil
LIGHT	very high; full, direct sun

WANTS Hot, bright, dry air like a desert shelf.
AVOID Damp, shady air rots the roots.

How to read a card

The thick black bar on each scale is the plant's happy range.

TEMP
10 to 40 °C.

HUMIDITY
0 to 100%.

LIGHT
shade on the left to full sun on the right. A plant in the wrong zone tells you how it suffers: too dry wilts, too bright scorches, too damp molds.

Control dial board

Set each dial for the plant you are placing, then record where each plant goes and the tour clue that backs it.

TEMPERATURE (°C)

10 20 30 40

mark your setting with a dry-erase pen

HUMIDITY (% RH)

0 25 50 75 100

mark your setting with a dry-erase pen

LIGHT

shade filtered bright full sun

mark your setting with a dry-erase pen

Placements: defend each one with a tour clue

PLANT	ZONE YOU CHOSE	TEMP	HUMIDITY	LIGHT	TOUR CLUE THAT SUPPORTS IT
Fern					
Moth orchid					
Cactus					

Tour clue sheet

As you tour, record what you see, feel, and hear for each zone: warm or cool, damp or dry, bright or shaded, plus any reading the docent points out. Back at the table this is your evidence for matching plants to zones.

GREENHOUSE ZONE	TEMPERATURE	HUMIDITY (DAMP/DRY)	LIGHT (SHADE/BRIGHT)	WHAT THE DOCENT SAID / EVIDENCE
Main greenhouse				
Hoop house				
Shaded bench				
Bright bench				

Tip: you do not need exact numbers. "Warmer and more humid than the hoop house" is good evidence. Use the control-system or weather-station readout if the docent shows one.

Pinecone Weather Machine

A humidity sensor with no battery and no code

At a glance: Build a humidity sensor with no battery and no code. About 80 min; 4 teams.

MATERIALS

- dry pine cones
- index cards
- wood skewers
- spray bottles
- rulers
- masking tape
- paper fasteners

FACILITATION ARC

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DEBRIEF

- Why does a two-layer strip bend when only one layer swells?
- What made your indicator faster?
- Where would a no-power humidity sensor be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Response accuracy	35
Response speed	20
Readable output scale	20
Biomimicry explanation	15
Redesign quality	10
Total	**100**

SOURCES AskNature Pine Cones Strategy

Pollinator Network Draft and Build

Design a habitat that works all season

At a glance: Draft a pollinator habitat that works all season, not just on the prettiest day. About 80 min; 4 teams.

MATERIALS

- plant cards
- pollinator cards
- grid boards
- stickers
- markers
- season tokens

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: native plants and clumping; cover spring, summer, and fall; change one placement at a time.
- Draft and place the plant and pollinator cards on the grid, then check the season coverage.
- Find a gap or an unfed pollinator, fix one placement, and recheck the coverage.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Where was the biggest gap in your bloom calendar?
- Why does clumping help pollinators?
- What happens to the network if one season is missing?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Seasonal coverage	30
Pollinator fit	25
Native and clumping logic	20
Layout clarity	15
Defense	10
Total	**100**

SOURCES USDA Pollinators; US Forest Service pollinator guidance

Plant cards

Each plant is native here. The dark season chips show when it blooms; the visitors are the pollinators it feeds. Place plants so something blooms in spring, summer, and fall, and group the same plant in clumps.

Serviceberry
Native

Amelanchier canadensis

BLOOMS

SPRING SUMMER FALL

VISITED BY bees, flies

One of the first trees to bloom each April.

Golden Alexanders
Native

Zizia aurea

BLOOMS

SPRING **SUMMER** FALL

VISITED BY bees, flies, beetles, butterflies

Tiny yellow flowers feed early short-tongued insects.

Eastern Redbud
Native

Cercis canadensis

BLOOMS

SPRING SUMMER FALL

VISITED BY bees, butterflies

Pink flowers bloom on bare branches in April.

Butterfly Weed
Native

Asclepias tuberosa

BLOOMS

SPRING **SUMMER** FALL

VISITED BY butterflies, bees, hummingbirds

Orange milkweed and a monarch caterpillar host.

Wild Bergamot
Native

Monarda fistulosa

BLOOMS

SPRING **SUMMER** FALL

VISITED BY bees, butterflies, hummingbirds, moths

Lavender pompoms that almost everyone visits.

Black-eyed Susan
Native

Rudbeckia hirta

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY bees, butterflies, flies

Golden petals with a dark center; a summer classic.

Mountain Mint
Native

Pycnanthemum tenuifolium

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY bees, flies, butterflies, beetles

Flat white flowers draw a huge mix of insects.

Summersweet
Native

Clethra alnifolia

BLOOMS

SPRING **SUMMER** FALL

VISITED BY bees, butterflies, hummingbirds

Sweet-smelling white spikes for shady, wet spots.

Cardinal Flower
Native

Lobelia cardinalis

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY hummingbirds, butterflies

Red tubes built for hummingbird beaks.

Joe-Pye Weed
Native

Eutrochium purpureum

BLOOMS

SPRING **SUMMER** **FALL**

VISITED BY butterflies, bees

Tall pink clouds buzzing in late summer.

New England Aster
Native

Symphyotrichum novae-angliae

BLOOMS

SPRING SUMMER **FALL**

VISITED BY bees, butterflies, flies

Purple fall blooms fuel migrating monarchs.

Goldenrod
Native

Solidago rugosa

BLOOMS

SPRING SUMMER **FALL**

VISITED BY bees, butterflies, flies, beetles

Yellow plumes blooming right up to frost.

Pollinator cards

Each pollinator needs flowers in bloom across every season it is active. Match it to plants whose flower type fits (hummingbirds to red tubes, flies and beetles to flat open clusters, bees and butterflies to most flowers).

Native Bees

ACTIVE

SPRING SUMMER FALL

VISITS Open daisy, mint, and clustered flowers with easy pollen.

Most are solitary and do not sting.

Bumble Bees

ACTIVE

SPRING SUMMER FALL

VISITS Tube and hooded flowers like bergamot they buzz open.

Queens wake first and need early spring blooms.

Butterflies and Monarchs

ACTIVE

SPRING SUMMER FALL

VISITS Flat-topped clusters and big landing-pad flowers.

Monarchs fuel on fall asters before migrating.

Hummingbirds

ACTIVE

SPRING SUMMER FALL

VISITS Red and pink tubular flowers full of nectar.

Long beaks reach where bees cannot.

Hoverflies and Flies

ACTIVE

SPRING SUMMER FALL

VISITS Shallow open flowers like mountain mint.

Hoverflies copy bees but cannot sting.

Beetles

ACTIVE

SPRING SUMMER FALL

VISITS Flat, sturdy flower clusters they can crawl across.

Some of Earth's oldest pollinators.

Grid board

Place your plant cards in the season they bloom (a plant that blooms twice can sit in two columns). Keep something in every column, and clump the same plant together.

SPRING	SUMMER	FALL

Pollinator food check

Tick a box only if a plant in that season feeds this pollinator. Every pollinator should have a tick in each season it is active.

POLLINATOR	SPRING	SUMMER	FALL
Native Bees			
Bumble Bees			
Butterflies and Monarchs			
Hummingbirds			
Hoverflies and Flies			
Beetles			

Arboretum Eco-Quest

Solve outdoor ecology checkpoints

At a glance: Read the living collection and solve the checkpoints before time runs out. About 80 min; 4 teams.

MATERIALS

- clue cards
- route map
- clipboards
- pencils
- evidence token bags
- optional QR codes

FACILITATION ARC

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- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	**100**

SOURCES Temple Ambler Arboretum; Arboretum Explorer

Minecraft Tree World Resilience Cup

Design a climate-resilient landscape

At a glance: Design a landscape that survives stress because of how it is built, not luck. About 80 min; 4 teams.

MATERIALS

- Minecraft Education devices and template or paper grid kit
- rubric cards
- display timer

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which feature would help most in a real storm?
- Why does diversity add resilience?
- What in your build would fail first, and how would you fix it?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Resilience features	30
Biodiversity support	25
Realism	20
Explanation	15
Build polish	10
Total	**100**

SOURCES Minecraft Education Biodiversity

Tree Ring Climate Detective

Read rings to reconstruct past climate

At a glance: Read the rings to reconstruct the climate events a tree lived through. About 80 min; 4 teams.

MATERIALS

- printed ring cards
- wood cookies or images
- answer boards
- timers
- claim-evidence cards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: read each card from center to bark and cite the exact rings for every inference.
- Read the ring sequences and infer the climate events.
- Back each claim with the exact rings on a claim-evidence card, then synthesize the story.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

SCORING (100 POINTS)

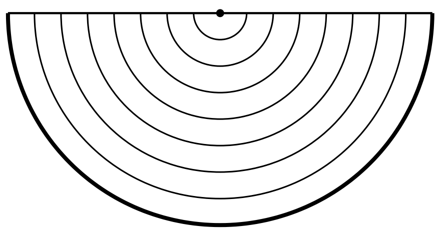
SCORING CRITERION	POINTS
Correct inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	**100**

SOURCES NOAA NCEI Tree Ring product and paleoclimatology (https://www.ncei.noaa.gov/products/paleoclimatology/tree-ring); NOAA Climate.gov, How tree rings tell time and climate history (https://www.climate.gov/news-features/blogs/beyond-data/how-tree-rings-tell-time-and-climate-history); UCAR Center for Science Education, Tree Rings and Climate (https://scied.ucar.edu/learning-zone/how-climate-works/tree-rings-and-climate).

Ring cards

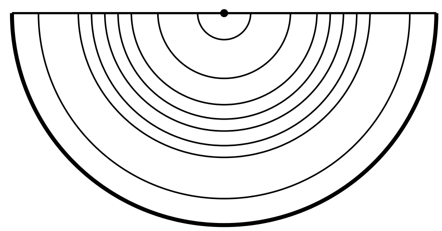
Each card is one tree's life. Read from the center (pith) out to the bark, one ring per year. Wide ring = a favorable year, narrow = a stress year (drought, cold, or crowding), a scar = a fire the tree lived through. A ring is a proxy, not a thermometer.

CARD A



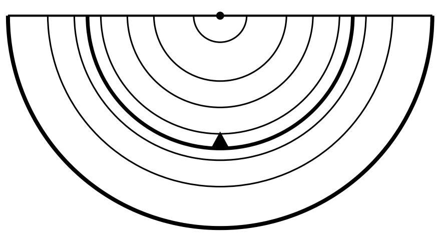
center (pith) on the flat edge, bark on the curve

CARD B



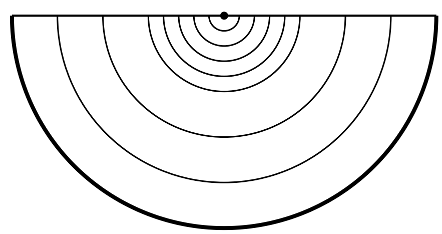
center (pith) on the flat edge, bark on the curve

CARD C



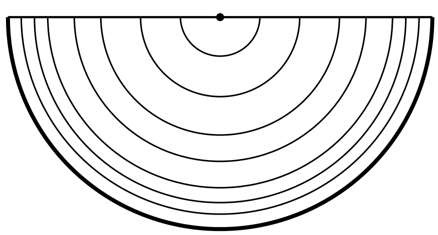
center (pith) on the flat edge, bark on the curve

CARD D



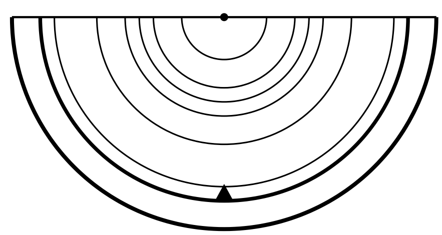
center (pith) on the flat edge, bark on the curve

CARD E



center (pith) on the flat edge, bark on the curve

CARD F



center (pith) on the flat edge, bark on the curve

Answer board

For each card, write what you see, the climate event you infer, and the exact rings that prove it. Read from the center out.

CARD	WHAT YOU SEE (WIDE / NARROW / SCAR)	EVENT YOU INFER	THE RINGS THAT PROVE IT
Card A			
Card B			
Card C			
Card D			
Card E			
Card F			

Claim-evidence cards

Use one card for each climate event you claim. Name the event, point to the exact rings, and explain why those rings prove it.

CLAIM, EVIDENCE, REASONING

Card: ____ Our claim (the event):

____

Evidence (which rings):

____

Reasoning (why those rings show it):

____

CLAIM, EVIDENCE, REASONING

Card: ____ Our claim (the event):

____

Evidence (which rings):

____

Reasoning (why those rings show it):

____

CLAIM, EVIDENCE, REASONING

Card: ____ Our claim (the event):

____

Evidence (which rings):

____

Reasoning (why those rings show it):

____

CLAIM, EVIDENCE, REASONING

Card: ____ Our claim (the event):

____

Evidence (which rings):

____

Reasoning (why those rings show it):

____

Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

At a glance: Build a surface that sheds water and dirt the way a lotus leaf does. About 80 min; 4 teams.

MATERIALS

- wax paper
- cardstock
- sandpaper strips
- masking tape
- pipettes
- water
- pepper or cocoa residue
- ramps

FACILITATION ARC

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- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	**100**

SOURCES AskNature Lotus Leaf Self-Cleaning Surface

Leaf Stomata Microscope Detective

Count stomata and rank leaves by water strategy

At a glance: Count the breathing pores and rank leaves by how they manage water. About 80 min; 4 teams.

MATERIALS

- leaf samples
- clear tape
- clear nail polish or prepared slides
- USB microscopes or magnifiers
- grid sheets
- gloves

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why count several fields instead of one?
- What does a high stomata count suggest about a leaf's habitat?
- Where might error sneak into your count?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Counting accuracy	30
Ranking evidence	25
Team data table	10
Cleanup and care	10
Total	**100**

SOURCES Leaf stomata microscopy adaptations; Science Buddies plant biology

BACKUP ACTIVITIES

Ready alternates

INSTRUCTOR · TTB-01 · FIELD SCIENCE

Root Grip Erosion Rescue

Can roots hold a slope together in a storm?

At a glance: Can roots hold a slope together during a storm? About 60 min; 4 teams.

MATERIALS

- Trays
- Soil
- String or yarn roots
- Craft sticks
- Watering cups

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why did the rooted slope hold more soil?
- How would deeper roots change the result?
- Where is this used to protect real hillsides?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	**100**

SOURCES EPA green infrastructure and erosion lessons

Tree Height Triangulation Shootout

Measure a tree without climbing it

At a glance: Measure a tree without climbing it. About 60 min; 4 teams.

MATERIALS

- Paper clinometers
- String and washers
- Rulers and tape measures
- Clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	**100**

SOURCES GLOBE Trees Activities

INSTRUCTOR · TTB-03 · DATA + CLIMATE

Urban Heat Shade Dash

Find the coolest route through a hot campus

At a glance: Find the coolest route through a hot campus. About 60 min; 4 teams.

MATERIALS

- Thermometers or IR thermometers
- Campus maps
- Clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How big was the sun-to-shade temperature gap?
- What surface was hottest, and why?
- How do cities use shade to cool streets?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Cool route design	35
Data quality	25
Reasoning	20
Communication	20
Total	100

SOURCES NASA MyNASAData Urban Heat Island

Photosynthesis Float-Off Playoffs

Make leaf disks float by making oxygen

At a glance: Make leaf disks float by producing oxygen. About 60 min; 4 teams.

MATERIALS

- Spinach leaves
- Baking soda solution
- Cups
- Oral syringes, no needle
- Timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does oxygen make the disks float?
- Which variable changed the rate most?
- How is this evidence of photosynthesis?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Controlled-variable design	25
Data table	20
Explanation	10
Cleanup	10
Total	**100**

SOURCES Exploratorium Photosynthetic Floatation